DISCRETE MATH · ELECTION THEORY

Voting, Power & Apportionment Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Voting, Power & Apportionment Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Schedule: 14 A > B > C, 10 C > B > A, 8 B > C > A, 3 C > A > B. Predict the plurality winner. Is it a majority?

2. In [6: 4, 3, 2], who do you think has the most power? Is it proportional to weight?

2 Explore use the tool

1. Enter the schedule from Predict 1. Record the winner under each method.

2. Is there a Condorcet winner? Which methods violate the Condorcet criterion here?

3. Weighted voting [6: 4, 3, 2]: list the winning coalitions and critical players.

4. Apportionment: 130, 690, 410 with Hamilton. Record seats for 13 and 14 seats. What happens?

3 Record use the tool and fill in the table

Weighted voting systems

System	Dictator?	Veto power	Dummy	Banzhaf counts
[6: 4, 3, 2]				
[4: 3, 2, 2]				
[6: 4, 2, 1]				
[5: 5, 2, 1]				

Apportion 10 seats: populations 450, 330, 220 (SD = 100)

Method	Divisor used	Seats
Hamilton		
Jefferson		
Adams		

4 Explain what did you discover?

1. A majority is _____ than half of the votes.
2. A player with weight at least the quota is a _____; a player who is never critical is a _____.
3. Hamilton always satisfies the _____ rule but can produce the Alabama paradox; divisor methods avoid the paradox but can violate the quota rule.
4. Use your Explore 1 results to explain why "the winner" of an election depends on the method, and why the method should be chosen before the votes are counted.

5 Apply a new situation, no tool

1. [10: 6, 5, 4]: find the Banzhaf power index of each player.
2. Is [7: 4, 3, 2, 1] a valid system? Does anyone have veto power?
3. Populations 5100, 1600, 1500, 1200; 20 seats; Jefferson. Find a divisor and the seats. Is the quota rule met?
4. Ann bids $300 for a house and $120 for a car; Ben bids $240 and $180 (sealed bids). Who gets what, and what cash changes hands?